THE CHINESE ECONOMY

A JOURNEY INTO HISTORY, ECONOMICS, AND POLICY

Lecture 5: From the hukou to the Lewis turning point

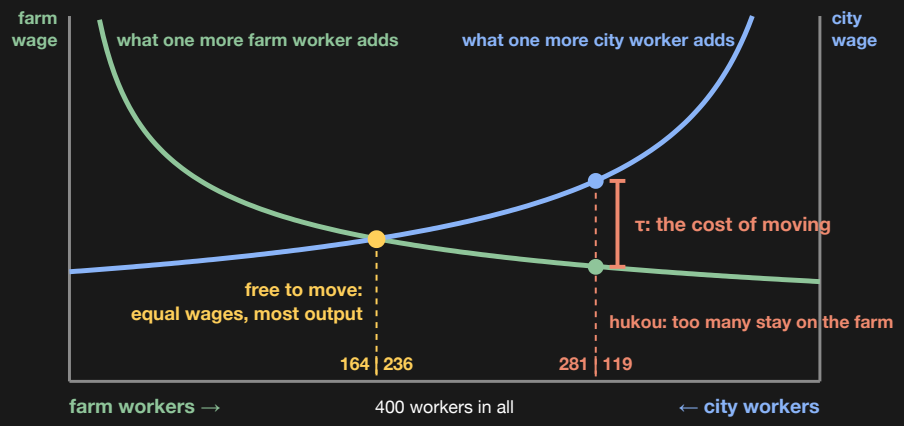
TODAY

- **Homework**: three questions, with the Hukou Wedge Lab on the screen
- **The story of our model**, and what it asks us to measure
- **The hukou changes**: from the 1980s, step by step
- **Two papers, told as stories**: Tombe and Zhu (2019), Khanna, Liang, Mobarak and Song (2025)
- Has China **run out of farm workers**?

HOMEWORK 1

HUKOU WEDGE LAB

Compare the farm wage and the city wage with and without the hukou. Who gains from the hukou, and who loses? Who might want to keep it?

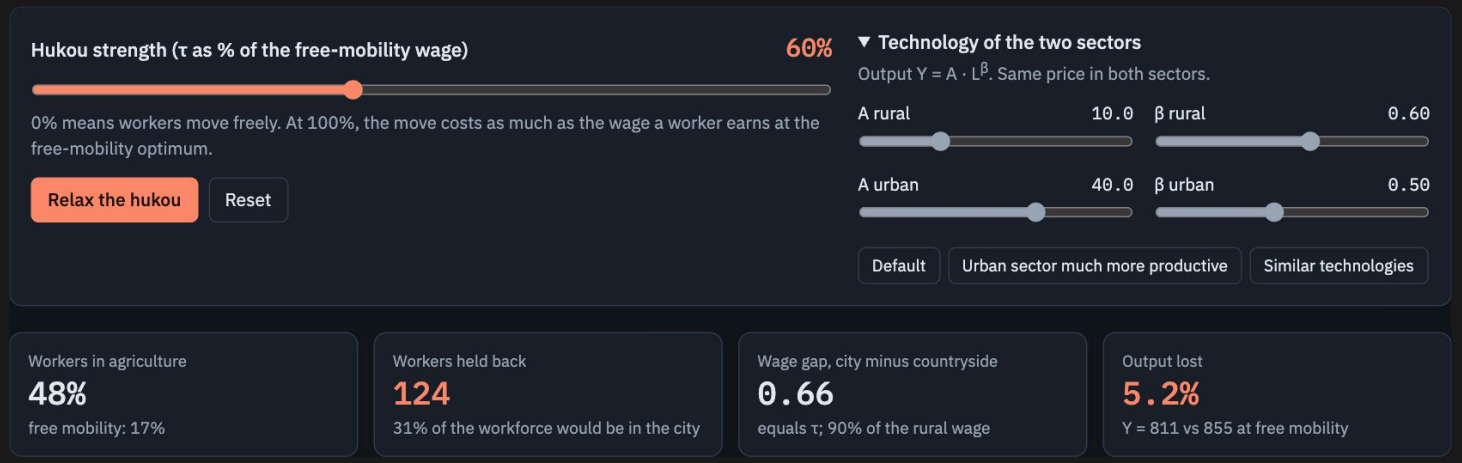


- Free to move: both wages **0.78**
- With the hukou: farm wage **0.63**, city wage **1.10**
- **Gains**: who is already in the city. **Loses**: who is kept on the farm. The economy loses about **4%** of output
- Who wants to keep it? Those who gain are the city dwellers, closer to the state and to the rations. A barrier with **winners** lasts

HOMEWORK 2

HUKOU WEDGE LAB

Press "Urban sector much more productive". With the same hukou, are more workers held back? Is more output lost? What does it say about the cost of the hukou as China's cities grew richer?



- **124** workers held back instead of 117; output lost **5.2%** instead of 4.4%
- The more the cities pull, the more the **same** barrier costs. After 1978 China's cities grew fast: keeping the hukou became more and more **expensive**

HOMEWORK 3

HUKOU WEDGE LAB

In Lewis's picture, one farm worker fewer costs almost nothing. Would the hukou cost more or less output than in our model? Why?

- **More**. In Lewis, a worker kept on the farm produces **almost nothing** there: we lose almost all he would have produced in the city
- In our model he still produces **something** on the farm: the loss is the **difference**
- So the hukou is most costly when the countryside is **full of extra hands**: China around 1980

THE STORY OF OUR MODEL

Question: what does a cost of moving do to where people work, to wages and to output?

Two sectors, farm F and city C, and L workers in all: $L_F + L_C = L$

Output in each sector:

$$Y_F = A_F \, L_F^{\gamma_F}$$

$$Y_C = A_C \, L_C^{\gamma_C}$$

in short $Y_X = A_X \, L_X^{\gamma_X}$, with $X \in \{F, C\}$ and $0 < \gamma_X < 1$: on the same land, or in the same factory, each extra worker adds less

What one more worker adds, the marginal product: $MPL_X = \gamma_X \, A_X \, L_X^{\gamma_X - 1}$. It falls as L_X grows, and rises with A_X

Wage and marginal product: in a market economy the wage equals the MPL. In a plan it need not, but total output still depends on the MPL. So in pre-market China the vertical axis of the two-origin diagram is the MPL, not the wage: read "wage" as MPL throughout. The output lost, the cost of the inefficiency, is measured correctly either way

Mechanism: free to move, $MPL_F = MPL_C$ and output is as large as possible. With the hukou, $MPL_C - MPL_F = \tau$: too many on the farm, output lost

RULES OF THE WORLD, AND WHAT IF ONE FAILS

The model assumes	If it fails
All workers are **alike**: no differences in education	Who moves depends on skills, and skills may pay differently in city and countryside. **Khanna et al.**, today
The **only** cost of moving is τ: no attachment to one's land, family, language	What we measure as τ includes them all. Keep it in mind for **Tombe and Zhu**
The wage equals the MPL	Measure the MPL instead. $MPL_X = \gamma_X A_X L_X^{\gamma_X - 1} = \gamma_X \, Y_X / L_X$: **output per worker** tells us the marginal product, and A_X is inside Y_X
A fixed number of workers, the same price for food and goods	More to model, same logic

QUESTIONS

THE STORY OF A MODEL

If the story is right, what should we see in the data?

- City wages **above** farm wages, for years, by about τ
- **Fewer** people moving than the wage gap alone would suggest
- When τ **falls**: more people move, the gap shrinks, output rises

How could we check it?

- We need τ to **change**, over time and across regions
- What should we measure?
    - who moves
    - their wages
    - the output
    - everything **before and after** the change
    - for **treated and untreated** regions

DOES THE HUKOU CHANGE?

- If τ never moved, we could not tell the barrier from everything else that makes cities richer
- **It moved.** From the 1980s, step by step, and not for everybody in the same way
- Two different things can change: being allowed to **move** (work elsewhere, keeping your hukou), and being allowed to **settle** (get the local hukou, its schools, its pension)

THE HUKOU CHANGES, 1980-2005

When	What	Move or settle?
1980s	Working outside one's hukou place is allowed; temporary hukou categories	Move
1984	Peasants may register in market towns, with "self-supplied grain"	Settle (small towns)
1992	Grain rationing ends almost everywhere: a city ration is no longer needed to live there	Move
1992-1998	Cities sell local hukou to investors and the wealthy, as the **blue-stamp** hukou: about **3 million** bought by 1993, some 8,000 yuan each, more than a year's urban wage. Graduates of good colleges move easily	Settle (money, skills)
1998	A child may take the status of **either** parent, no longer only the mother's	Settle (family)
2003	Sun Zhigang dies in police detention in Guangzhou, without a temporary permit; many coastal provinces drop the permit	Move
2005	The central government acknowledges that migrants have a right to live and work in the city; most cities register them temporarily	Move



A Guangzhou blue-stamp hukou booklet, 1998. Wikimedia Commons, public domain

Naughton (2007), ch. 5, section 5.1.5 "Invisible Walls", pp. 124-126; Chan and Buckingham (2008); Tombe and Zhu (2019), p. 6

POINTS, AND THE SIZE OF THE CITY, 2009-2016

When	What	Move or settle?
2009	**Zhongshan** (Guangdong) pilots a **points system** for migrants; Shanghai: permit holders may get the hukou after 7 years, with skills	**Settle (points, skills)**
2010	Guangdong: points for rural migrant workers, 60 points to apply	**Settle (points)**
2013	Shanghai: points give access to public **services**, not to the hukou	**Services**
2014	State Council: open fully in towns and small cities; stricter the larger the city; cities above **5 million** keep points. The rural/urban label is abolished. Target: 100 million people settled by 2020	**Settle (by city size)**
2016	Residence permit after six months: schooling, jobs, health services without the hukou	**Move + services**

Settling opened from the **bottom** of the ladder. The cities people most want to reach chose **whom** to let in

State Council documents of 2014, 2015 and 2016; Guangdong and Shanghai government texts; Wan and Vickers (2022). Naughton (2007) stops around 2006

HOW TO TELL A PAPER

1. **Motivation**: start from the **title**. What puzzle or fact makes the question worth asking?
2. **Question**: one sentence, generally with a cause and an effect
3. **Data**: what is measured, where, when. And: **could the data be wrong?** If they were, what would it do to the answer?
4. **Empirical strategy**: the comparison, and the **rules of the world** we must assume for it to give the causal effect
5. **Results**: the numbers, with their size in words
6. **Robustness**: are we sure the rules of the world hold in this case? How do the authors test alternative explanations, and possible biases, of what they find?
7. **Why it matters**: for China, for policy, for what we thought we knew

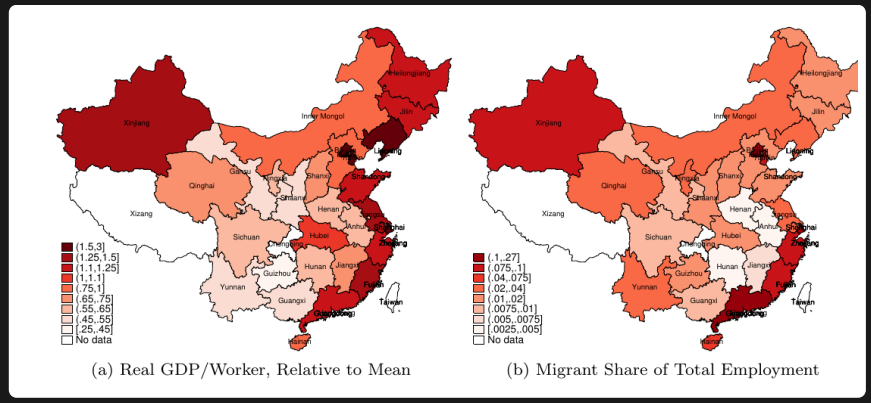
TRADE, MIGRATION, AND PRODUCTIVITY: A QUANTITATIVE ANALYSIS OF CHINA

Trevor Tombe, Xiaodong Zhu

American Economic Review, 2019

MOTIVATION

- The title names three things: **trade**, **migration**, **productivity**
- Around 2000 China had walls **inside** the country: for goods between provinces, and for people, the hukou
- Output per worker: richest five provinces against poorest five, **4 to 1**. Inside a province, city against farm, again **4 to 1**
- And the rich coast is where the migrants already are



Tombe and Zhu, Fig. 1: (a) real GDP per worker relative to the mean, (b) migrants as a share of employment, 2000

QUESTION

- How high were the costs of moving **goods** and **people** inside China?
- How much did they **fall** between 2000 and 2005?
- How much of the growth of output per worker came from that fall?

DATA

TOMBE AND ZHU (2019) · 3. DATA

- The population **census** of 2000 (1% sample) and the **mini-census** of 2005: who works outside the province, or the sector, of their hukou
- Migrants between provinces: **26.5 million** in 2000 (4.2% of workers), **49.0 million** in 2005 (7.2%). From farm to city inside a province: 90.1 and 120.4 million
- Output per worker by province and sector, from the statistical yearbooks; who buys from whom, from input-output tables
- 30 provinces, each split into farm and non-farm: 60 places to live and work

QUESTION

TOMBE AND ZHU (2019) · 3. DATA, AND WHAT IF THEY ARE WRONG

What if the census did not see all the migrants?

- The cost of moving is not observed: it is read off **how many people move**, given the income gap. Fewer movers counted means a **higher** measured cost
- Who could be missing: seasonal workers; people hiding from the police (before 2003 a migrant without a permit risked **deportation**); migrants who got the local hukou, counted as **locals**
- If the same share is missing in 2000 and in 2005, it matters little: the results use the **change**, and a constant bias cancels
- If more people hid in 2000 than in 2005, the **fall** in the cost looks larger than it was. People missing from the data: **sample selection**

EMPIRICAL STRATEGY

- We cannot **see** the cost of moving. We read it from what people **choose**, as with revealed preferences in micro
- For every pair of places: how much more one earns **there** than **here**, and how many people **actually move**
- A big gap and few movers: moving must be **very costly**. A small gap and many movers: it is cheap
- Part of the cost is real: a migrant **loses the income** of the village land. And people differ: some would move for little, some for no money at all
- Goods the same way: provinces that buy mostly **from themselves** face high costs of trading
- We compare **2000 with 2005**: what is fixed cancels
- What **changes** over time (highways, amenities) stays in the measured fall

RULES OF THE WORLD

- Each worker picks the province and sector with the highest real income **net of the cost**, with personal tastes on top
- The cost is paid **every year** one lives without the local hukou; migrants **lose their share** of village land income
- Provinces do not differ in how pleasant they are
- So everything else is counted as **cost**, including the attachment to one's land

FROM THE MODEL TO THE NUMBERS

TOMBE AND ZHU (2019) · 4. EMPIRICAL STRATEGY

Data

Who works where, by hukou: censuses of 2000 and 2005. Real income by province and sector: statistical yearbooks

→

Model

Each worker picks the place with the highest real income, minus the cost of moving. Tastes differ: how much is the one number estimated by a regression (about 1.5)

→

What comes out

The cost of moving for every pair of places, in 2000 and in 2005: the number that makes the model match the data

→

Counterfactual

Change only the costs, keep everything else: how much more output per worker?

- There is **no dependent variable** of a regression here: the cost is **computed**, not estimated
- Next two slides: what comes out (the cost of moving), then the counterfactual (where the growth came from)

The model's equation (additional material, end of the deck)

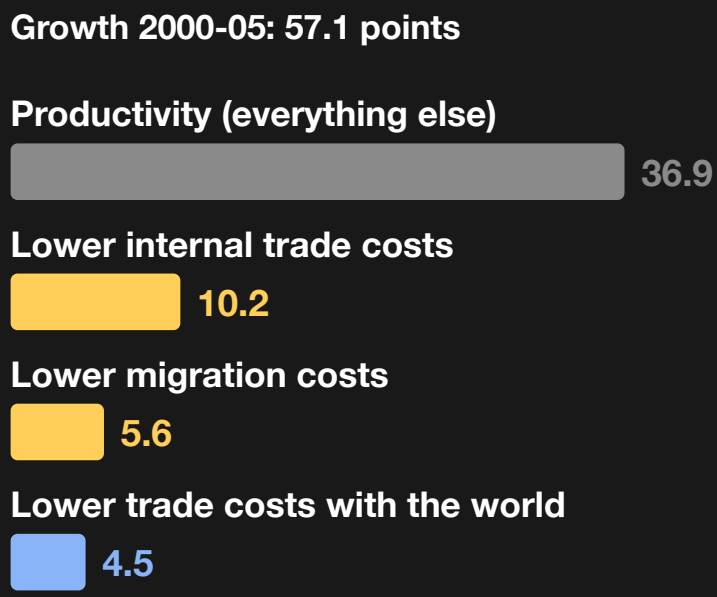
RESULTS: THE COST OF MOVING

From farm to city...	2000	2005	Change
inside one's province	2.21	1.83	-17%
to another province	25.21	15.43	**-39%**
all moves, on average	2.82		-18%

- 2.21 means: moving to a city of one's province was like **dividing** one's real income by 2.2, keeping about 45%
- To another province: by 25. The wall was high, and it was coming down

RESULTS: WHERE THE GROWTH CAME FROM

- **The question now**: output per worker grew by 57 points in 2000-05. How much of it came from lower costs of moving and trading?
- **What they do**: run the model again letting **one** cost change at a time, everything else as it was
- **Reading**: lower walls **inside** China (trade 10.2 + migration 5.6) give about **28%** of the growth; with the walls at the border, **36%**
- **What is "productivity"**: everything else. Better technology, more capital per worker, and how capital is spread across **firms** within each province and sector. That last part is **lecture 6**



ROBUSTNESS

TOMBE AND ZHU (2019) · 6. ROBUSTNESS

- A rule of the world: provinces do not differ in how pleasant they are. If the poor ones were nicer, the cost would look **too high**. Harmless if it did not change between 2000 and 2005: they use **changes**
- How strongly people react to income: **estimated** from the data, between 1.2 and 1.6 (they use 1.5), not assumed
- Does the measured cost behave like a cost? It rises with **distance** and with **age**; migrants earn about **four times** what they think they would earn at home
- Is it the hukou? The cost fell more for migrants to the five provinces that **reformed** their hukou (22% against 15%). But **highways** were built too: the authors name them as another reason

WHY IT MATTERS

- China's growth in these years is usually called **export-led** (the WTO, 2001). Here the walls **inside** the country mattered more
- The walls were still high in 2005. If one worker in three lived outside their home province, as in the United States: output per worker **+12.8%**, welfare **+45.6%**
- The τ of last week is **real**, it is **large**, and it **fell**

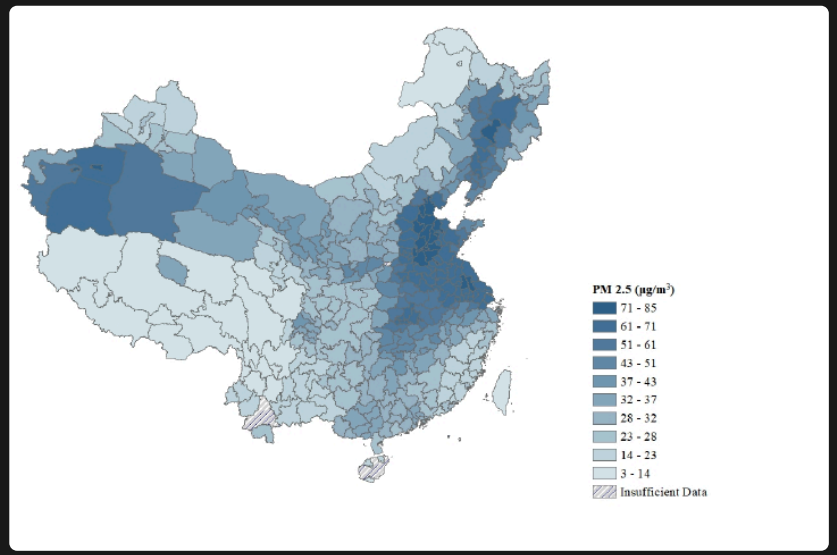
THE PRODUCTIVITY CONSEQUENCES OF POLLUTION-INDUCED MIGRATION IN CHINA

Gaurav Khanna, Wenquan Liang, Ahmed Mushfiq Mobarak, Ran Song

American Economic Journal: Applied Economics, 2025

MOTIVATION

- China grew with two features: **mass migration** and **severe air pollution**
- In 2015 fine particles (PM2.5) were above the WHO guideline in **96%** of cities, on average **four times** the safe level
- Pollution rose a lot in **some** cities only: a worker can escape it by **moving**
- And who leaves changes what a city can produce



Khanna et al., Fig. 3B: PM2.5 concentration, 2015 (satellite data)

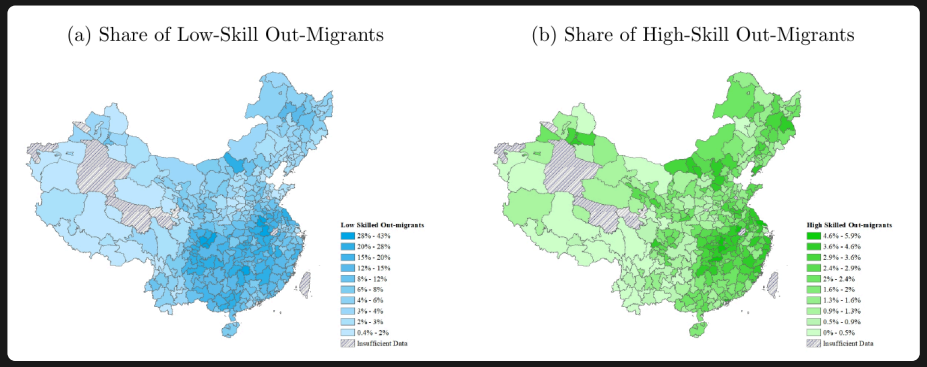
QUESTION

- Do workers **leave** polluted cities?
- Do the **skilled** leave more than the others?
- And what does this sorting cost the **national** economy?

DATA

KHANNA, LIANG, MOBARAK AND SONG (2025) · 3. DATA

- **2015** one-percent census: 761,548 working-age people from 332 cities. A migrant is someone living away from their **hukou city** for more than six months
- Skilled: some college or more (16%)
- Pollution measured from **satellites**, not from the official monitors
- A **hukou index** by city and skill: how hard it is to get the local hukou. Hardest in **Beijing**, then the other big cities



Khanna et al., Fig. 4: share of out-migrants by hukou city, (A) low-skill, (B) high-skill

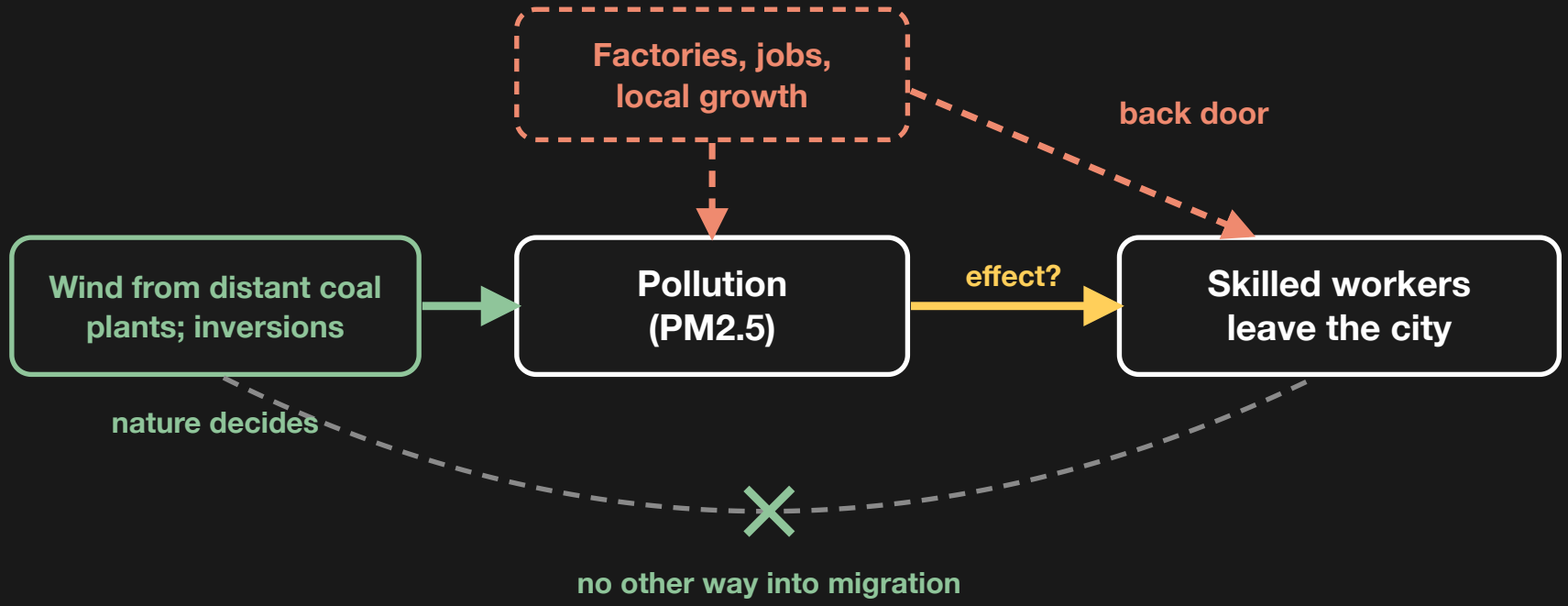
QUESTION

KHANNA, LIANG, MOBARAK AND SONG (2025) · 3. DATA, AND WHAT IF THEY ARE WRONG

Why not the official pollution data? And who could be missing from the census?

- The authors: official air-quality data "may be subject to **manipulation** by local governments"
- If a city's officials under-report pollution when they push for growth, and growth also keeps workers: the error moves with the outcome. A **confounder**. Satellites avoid it
- Who is missing: someone who left **and got the local hukou** counts as a local. Skilled people get it more easily: their departures are **undercounted**
- That is **sample selection** pushing the result **towards zero**: the true effect is, if anything, larger

EMPIRICAL STRATEGY

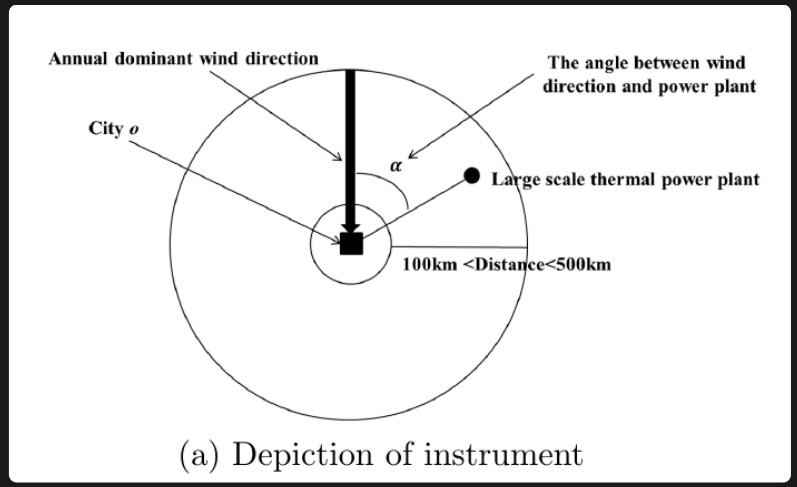


- Polluted cities have more **factories**, which also decide who stays: a simple comparison mixes the two
- So they use only the part of pollution that **nature** assigns: two lotteries

LOTTERY 1: THE WIND

KHANNA, LIANG, MOBARAK AND SONG (2025) · 4. EMPIRICAL STRATEGY

- Big coal plants **100 to 500 km** away, counted more when the prevailing wind blows **from** them towards the city
- **Rule of the world**: a distant plant affects a city's migration **only through the air** it sends. No local jobs, no local electricity

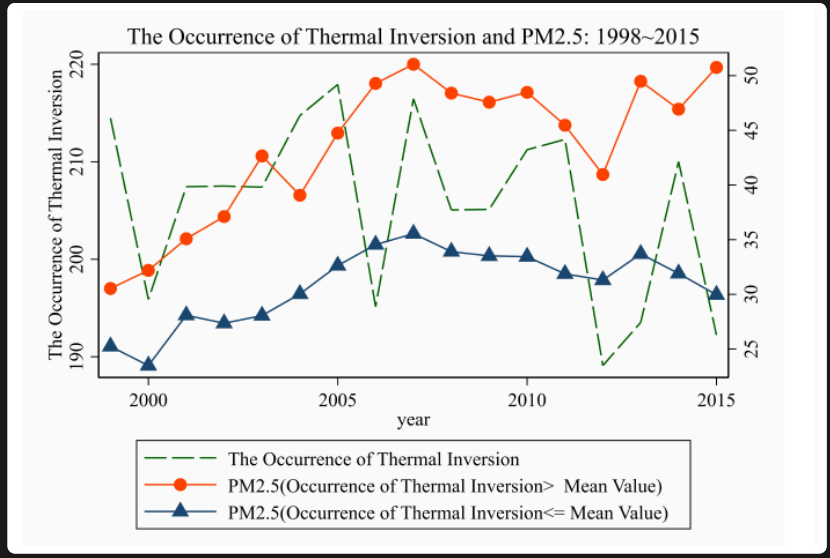


Khanna et al., Fig. 1A: the instrument. City o, the prevailing wind, a plant 100-500 km away, the angle α between them

LOTTERY 2: THERMAL INVERSIONS

KHANNA, LIANG, MOBARAK AND SONG (2025) · 4. EMPIRICAL STRATEGY

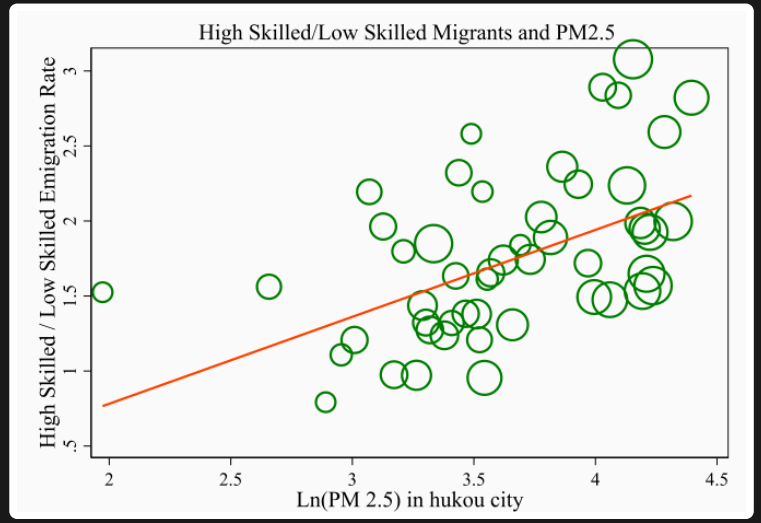
- Some days warm air sits **on top of** cold air and traps the smog over a city
- **Rule of the world**: inversions change migration **only by trapping pollution**, not as unpleasant weather in themselves; and pollution does not cause them



Khanna et al., Fig. 2 (left): inversions (green) and PM2.5 in cities with many (red) and few (blue) inversions, 1998-2015

RESULTS: WHO LEAVES

- Raw data: the dirtier the hukou city, the more skilled people leave **relative** to the others
- With the lotteries: **10%** more pollution raises the share of skilled people who leave by **1.4 points** (about one in ten leave on average). For the others: **0.6**, not distinguishable from zero
- Following the same people over the years, and counting **every** move (not only a change of hukou city): the **same pattern**
- The skilled are also the most sensitive to **hukou rules**, though the rules favour them



Khanna et al., Fig. 5A: ratio of high- to low-skilled out-migration against PM2.5 in the hukou city, 2015

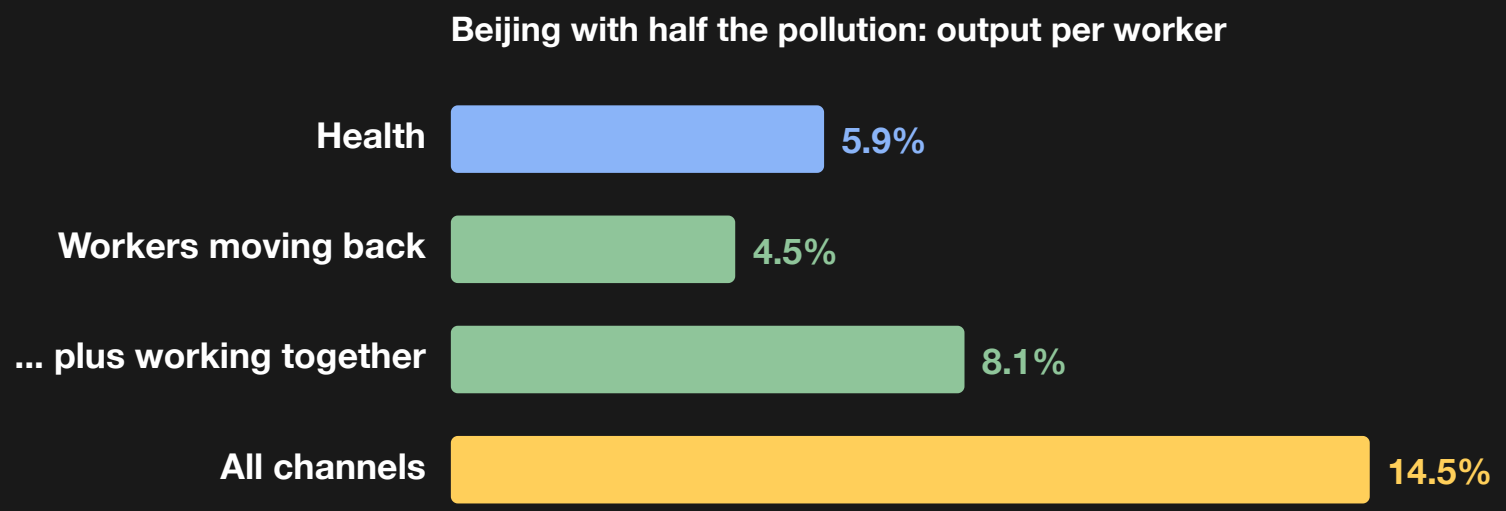
WHO MAY SETTLE: POINTS

KHANNA, LIANG, MOBARAK AND SONG (2025) · CONTEXT: WHY THE BARRIER DIFFERS BY SKILL

City	Points needed	Bachelor's degree	Doctorate
Beijing	varies	15	37
Shanghai	72	21	27
Shenzhen	100	80	100

- The relaxation was **selective**: in the big cities, a degree buys points towards the local hukou
- Before 1979 the state chose by its **needs**. Now the city chooses by **skills** and money

RESULTS: WHAT CLEANER AIR IS WORTH



- Bringing skilled workers back is worth about as much as the **health** gain. Wages of the unskilled rise **18.5%**, of the skilled only slightly

ROBUSTNESS

- **The wind**: big plants are placed by the **central** government; the result holds without nearby plants, with old plants only, without the coal regions; wind turned by 90° or 180° predicts much less
- **The inversions**: no trend over time, yet cities with many of them saw pollution rise faster; past pollution does not predict them; temperature, wind, sun and humidity held fixed
- **Other data**: with the official air-quality index instead of satellites, the skilled still leave
- **Same people over time**: following individuals across years, and counting every move, gives the same pattern

WHY IT MATTERS

- Ignore that people **move**, and you underestimate what pollution costs
- **Pollution policy** and **hukou policy** are tied: the largest gains come from cleaner air **and** easier hukou together
- Points favour the educated: the **unskilled** may stay trapped in polluted, low-wage cities while the skilled leave
- For our model: workers are **not alike**, and the wedge is not the same for everybody

HAS CHINA RUN OUT OF FARM WORKERS?

	The evidence
Yes Zhang, Yang, Wang (2011); Cai and Du (2011)	Real wages of hired workers in villages were flat for ten years, then rose from **2003**, even in the slack season. In poor Gansu villages: 12.8 yuan a day (1993), 12.9 (2003), 19.6 (2006). Migrant wages rose too
No Golley and Meng (2011)	Migrant earnings rose about **30%** in 2000-09, against **93%** for urban workers. Half the farm workforce works **part time**. Barriers keep people in the villages and shorten their stays in the cities: with other rules, 150 million migrants could become 300
Not yet Das and N'Diaye (IMF, 2013)	About **150 million** extra workers in 2010, 33 million in 2020: the turning point between **2020 and 2025**. Faster hukou reform pushes it later, to 2025-2030

QUESTION

THE LEWIS TURNING POINT

City wages are rising fast. Is that proof that the extra farm workers are gone?

- **Not by itself.** In our model, a high τ keeps workers on the farm **and** keeps city wages high, at the same time
- "Gone" and "**blocked**" look alike in city wages. To tell them apart, look at the **countryside**: wages in the slack season, days actually worked on the farm, how long migrants stay
- That is why the two sides use different data: village wages for the yes camp, days worked and length of stay for the no camp

NEXT WEEK: INSIDE THE SECTORS

- Moving workers from the farm to the city is **half** of the story
- Inside each sector, some firms are much more productive than others. Moving workers and capital **between firms** can raise output too
- In China the sector with the largest gap was the state's own: **state-owned enterprises**

HOMEWORK FOR NEXT WEEK

- Tell the story of **Tombe and Zhu** or of **Khanna et al.** in the seven steps: motivation, question, data (and what if they are wrong), strategy (and its rules of the world), results, robustness, why it matters. One page at most
- Pick **one rule of the world** of our model and say how the story changes if it fails
- Not graded. Send it to francesco.manaresi@bancaditalia.it

ADDITIONAL MATERIAL: THE MODEL'S EQUATION

TOMBE AND ZHU (2019), EQUATION (10) AND SECTION III.A

Share of the workers registered in province n, sector j, who work in province i, sector k:

$$m_{ni}^{jk} = \frac{\left(V_i^k \, \delta_{ni}^{jk} / \mu_{ni}^{jk}\right)^{\kappa}}{\sum_{k'} \sum_{i'} \left(V_{i'}^{k'} \, \delta_{ni'}^{jk'} / \mu_{ni'}^{jk'}\right)^{\kappa}}$$

Compare movers with stayers and solve for the cost:

$$\mu_{ni}^{jk} = \frac{1}{\delta_{nn}^{jj}} \cdot \frac{V_i^k}{V_n^j} \cdot \left(\frac{m_{nn}^{jj}}{m_{ni}^{jk}}\right)^{1/\kappa}$$

- V: real income per worker in each place; m: who stays (m_{nn}^{jj}) and who moves (m_{ni}^{jk}), from the censuses
- δ_{nn}^{jj}: the village land income that stayers keep and migrants lose (about 1.9 on average)
- κ: how much tastes for places differ across people, estimated at about 1.5
- A big income ratio and few movers: a big μ